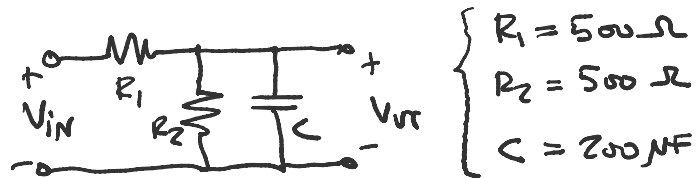
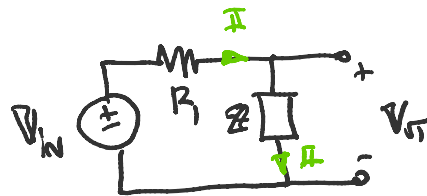
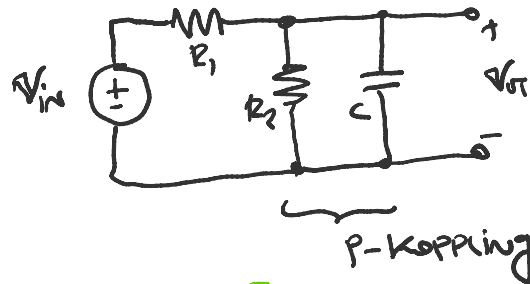


6.2 a)

Bestäm V_{out}/V_{in} för kretsen ovan.

Lösning:



KSPL ger

$$\begin{cases} -V_{in} + (R_1 + Z)I = 0 & (1) \\ -V_{out} + ZI = 0 & (2) \end{cases}$$

(2) i (1)

$$-V_{in} + (R_1 + Z) \frac{V_{out}}{Z} = 0$$

$$\frac{V_{out}}{V_{in}} = \frac{Z}{R_1 + Z}$$

(3)

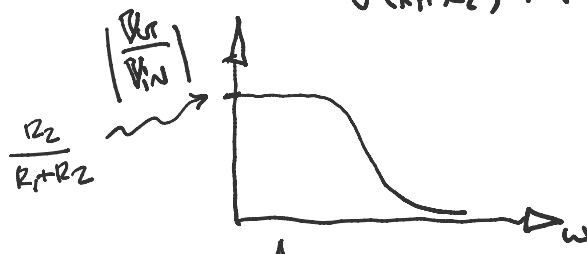
$$\text{MEN } Z^{-1} = \frac{1}{R_2} + j\omega C \Rightarrow Z = \frac{R_2}{1 + j\omega C R_2} \quad (4)$$

(4) i (3)

$$\frac{V_{out}}{V_{in}} = \frac{\frac{R_2}{1 + j\omega C R_2}}{R_1 + \frac{R_2}{1 + j\omega C R_2}} = \frac{R_2}{R_1 + R_2 + j\omega C R_2 R_1}$$

Tag Betspelt

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{R_2}{\sqrt{(R_1 + R_2)^2 + (\omega C R_2 R_1)^2}}$$



$R_1 + R_2$



SVAR:

$$\frac{V_{out}}{V_{in}} = \frac{R_2}{R_1 + R_2 + j\omega C R_1 R_2}$$